

A2 Level CS — Semester 1 Weeks 1–9: Advanced Data Representation & Communication | Victoria Park Academy

Victoria Park Academy

A2 Level Computer Science — Semester 1 Weeks 1–9

Advanced Data Representation & Advanced Communication

Weeks 1 – 9: Advanced Data Representation & Advanced Communication

Unit 1: Advanced Data Representation (Weeks 1–4)

Unit 2: Advanced Communication (Weeks 5–8)

Week 9: Consolidation & Formative Assessment

高级数据表示与通信 / Advanced Data Representation & Communication

本单元涵盖布尔代数、卡诺图、浮点数表示、TCP/IP协议栈、路由算法和DNS。This unit covers Boolean algebra, Karnaugh maps, floating-point representation, TCP/IP stack, routing algorithms, and DNS.

Week 1: Boolean Algebra — Laws and Theorems

Week 1: Boolean Algebra Foundations

Learning Objectives

Recall and apply the fundamental laws of Boolean algebra (identity, null, idempotent, inverse, commutative, associative, distributive, absorption)

Apply De Morgan's laws to simplify expressions involving complements of sums and products

Simplify Boolean expressions using algebraic manipulation

Prove equivalence of two Boolean expressions using truth tables and algebraic methods

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): Quick-fire truth table completion for AND, OR, NOT gates. Students use mini-whiteboards.

Direct Instruction (25 min): Teacher presents each Boolean law with a numerical analogy (e.g., distributive law: $3 \times (4 + 5) = 3 \times 4 + 3 \times 5$). Each law demonstrated with a truth table proof.

Worked Example (15 min): Simplify $A \cdot (A + B)$ using absorption law. Verify with truth table.

Guided Practice (25 min): Pairs complete scaffolded worksheet with 8 expressions to simplify. Bronze: 2-variable; Silver: 3-variable; Gold: expressions requiring multiple laws.

Plenary (5 min): Exit ticket: Simplify $X + X' \cdot Y$ without assistance.

Lesson 2 (2 hours):

Starter (10 min): Recall quiz on laws from previous lesson.

Direct Instruction (20 min): De Morgan's laws introduced with Venn diagram visualisation. Emphasise "break the bar, change the sign" mnemonic.

Worked Example (15 min): Simplify $(A + B + C)'$ step-by-step using De Morgan's.

Guided Practice (30 min): Students work in groups of 4. Each group assigned one complex expression to simplify, then present solution to class using visualiser.

Independent Practice (35 min): Past paper question set (A-Level Paper 1, Q3–5 from 2018–2023). Students select bronze (3 questions), silver (5 questions), or gold (7 questions + extension proof).

Plenary (10 min): Class discussion: "Why is Boolean algebra essential for computer hardware design?"

Resources

Worksheet: Boolean Algebra Laws — Scaffolded Practice (PDF)

Interactive truth table generator: <https://web.stanford.edu/class/cs103/tools/truth-table-tool/>

Physical logic gate kits for demonstration (if available)

Past paper compilation: Boolean simplification questions 2018–2023

Assessment Points

Formative: Exit ticket (Lesson 1) — teacher reviews before Lesson 2

Formative: Group presentation rubric (Lesson 2) — peer and teacher assessment

Homework: Simplify 5 expressions; prove equivalence using both truth table and algebraic methods

Bronze: Simplify expressions using single laws; complete truth tables with guidance.

Silver: Apply multiple laws in sequence; verify results independently.

Gold: Prove complex equivalences; derive simplified forms from word problems; explain why particular laws apply.

Week 2: Karnaugh Maps

Week 2: Karnaugh Maps (K-Maps)

Learning Objectives

Construct 3-variable and 4-variable Karnaugh maps from truth tables or Boolean expressions

Identify and group adjacent cells following grouping rules (groups of 1, 2, 4, 8, 16)

Derive minimal Sum-of-Products (SOP) and Product-of-Sums (POS) expressions

Recognise and handle "don't care" conditions in K-map simplification

Compare K-map method with algebraic simplification for efficiency and accuracy

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): Review simplified expression from Week 1 exit ticket. Ask: "Could we find this answer without algebra?"

Direct Instruction (30 min): Introduce K-map grid structure. Explain Gray code ordering (00, 01, 11, 10). Show why adjacency matters. Demonstrate 3-variable K-map construction from truth table.

Worked Example (20 min): Teacher completes a 3-variable K-map step-by-step: populate cells, identify groups, derive minimal expression. Show both SOP and POS approaches.

Guided Practice (30 min): Students receive K-map templates and work in pairs on 3 problems. Teacher circulates with targeted questions: "Why did you group those cells?" "Is there a larger group possible?"

Plenary (10 min): Whole-class check of solutions. Common errors addressed (wrapping around edges, diagonal grouping).

Lesson 2 (2 hours):

Starter (10 min): Quick K-map completion from expression: $F(A, B, C) = \Sigma(0, 2, 4, 6)$

Direct Instruction (20 min): 4-variable K-maps introduced. Emphasise wrap-around grouping (top-bottom, left-right, corners). Show "don't care" notation (X) and how it expands grouping possibilities.

Worked Example (15 min): 4-variable K-map with don't cares. Teacher thinks aloud: "This X lets me make a group of 8 instead of two groups of 4."

Guided Practice (35 min): Carousel activity — 6 stations around room, each with a different K-map problem. Students rotate every 6 minutes, building speed and confidence.

Independent Practice (30 min): Bronze: 3 three-variable problems; Silver: 2 four-variable problems; Gold: 1 four-variable with don't cares + design challenge: "A security system has 4 sensors. Use a K-map to minimise the alarm logic."

Plenary (10 min): Student reflection: "When would I prefer K-maps over algebra? When would algebra be faster?"

Resources

Printed K-map templates (3-variable and 4-variable) — one per student

Online K-map solver for verification: <https://www.charlie-coleman.com/kmaps/>

Logisim circuit files for verifying simplified expressions

Video: "Karnaugh Maps — Introduction" (Neso Academy, 15 min)

Assessment Points

Formative: Carousel station answers checked by teacher during rotation

Formative: Exit ticket — 4-variable K-map with one don't care

Homework: Complete 5 K-map problems; verify one using Logisim

Bronze: Complete 3-variable K-maps with teacher guidance; identify obvious groups.

Silver: Complete 4-variable K-maps independently; handle simple don't care conditions.

Gold: Solve complex 4-variable problems with multiple don't cares; design K-map from word problem; explain grouping choices rigorously.

Week 3: Floating-Point Representation

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Learning Objectives

Explain the structure of floating-point numbers: sign bit, exponent, mantissa (significand)

Convert denary numbers to IEEE 754 single-precision binary representation

Convert IEEE 754 single-precision binary back to denary

Explain the trade-off between precision and range in floating-point representation

Identify special values: zero, infinity, NaN (Not a Number)

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): "How would you store 0.1 in binary?" Discussion of the limitations of fixed-point representation.

Direct Instruction (30 min): Introduce scientific notation as analogy (e.g., 6.022×10^{23}). Extend to binary: 1.101×2^4 . Break down IEEE 754 single-precision format: 1 sign bit, 8 exponent bits (with bias 127), 23 mantissa bits. Explain hidden bit convention.

Worked Example (25 min): Convert +6.625 to IEEE 754 single-precision. Step-by-step: convert integer part, convert fractional part, normalise, determine exponent with bias, assemble bits.

Guided Practice (25 min): Pairs work on converting +9.25 and -3.5. Teacher provides conversion checklist: (1) Integer to binary, (2) Fraction to binary, (3) Normalise, (4) Apply bias, (5) Assemble.

Plenary (10 min): Whole-class verification using online converter. Discuss any discrepancies.

Lesson 2 (2 hours):

Starter (10 min): Reverse conversion starter: given binary pattern, what is the denary value?

Direct Instruction (20 min): Reverse conversion process: extract sign, exponent (subtract bias), mantissa (add hidden bit), calculate value. Special cases: exponent all zeros (denormalised), all ones (infinity/NaN).

Worked Example (15 min): Convert IEEE 754 pattern 0x40400000 to denary. Teacher demonstrates hex-to-binary breakdown.

Guided Practice (30 min): Students complete reverse conversion worksheet with 6 problems. Bronze: positive numbers only; Silver: include negative numbers; Gold: include special cases (infinity, zero).

Independent Practice (30 min): Past paper questions on floating-point from 2019–2023. Students self-select difficulty level.

Plenary (15 min): Discussion: "Why can't computers store 0.1 exactly? What are the implications for financial software?"

Resources

IEEE 754 conversion worksheets (printed)

Online converter for verification: <https://www.h-schmidt.net/FloatConverter/IEEE754.html>

Python demonstration: `struct.pack('!f', 0.1)` to show binary representation

Video: "How floating-point works" (Computerphile)

Assessment Points

Summative: Quiz 1 (Week 3) — Boolean & Floating-Point (20 marks)

Formative: Reverse conversion worksheet marked and returned next lesson

Homework: Convert 5 denary numbers to IEEE 754; explain why $0.1 + 0.2 \neq 0.3$ in floating-point

Bronze: Convert simple positive denary numbers to floating-point with structured worksheet.

Silver: Convert positive and negative numbers; perform reverse conversion accurately.

Gold: Handle special cases; explain precision limitations; discuss real-world implications (financial computing, scientific simulation).

Week 4: Normalisation & Errors

Week 4: Normalisation & Truncation Errors

Learning Objectives

Normalise floating-point numbers to maximise precision

Calculate absolute and relative error in floating-point representation

Explain truncation error and rounding error with examples

Discuss underflow and overflow conditions

Evaluate the suitability of floating-point for different applications

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): Compare two representations of the same number: 0.011×2^4 vs 1.1×2^2 . Which is better? Why?

Direct Instruction (25 min): Define normalisation: adjusting mantissa and exponent so the mantissa starts with 1 (for positive) or 0.1 (for signed). Show how normalisation maximises significant digits. Demonstrate normalising both positive and negative numbers.

Worked Example (15 min): Normalise $0.001101 \times 2^5 \rightarrow 1.101 \times 2^2$. Show the effect on precision.

Guided Practice (30 min): Students normalise 8 numbers (mix of positive, negative, already normalised, and denormalised). Checklist approach: "Is the mantissa in the correct range? Did I adjust the exponent correctly?"

Plenary (10 min): Exit ticket: Normalise 0.000111×2^6 and explain why this form is preferred.

Lesson 2 (2 hours):

Starter (10 min): Recall normalisation from previous lesson. Quick practice: is 1.01×2^3 normalised? Is 0.11×2^4 ?

Direct Instruction (25 min): Types of error: absolute error = $|\text{exact} - \text{approximate}|$; relative error = $\text{absolute error} / |\text{exact}|$. Truncation error: discarding less significant bits. Rounding error: adjusting last kept bit. Work through numerical examples.

Worked Example (15 min): Represent $1/3$ in 4-bit mantissa. Calculate absolute and relative error. Discuss how error accumulates in iterative calculations.

Guided Practice (25 min): Problem set: calculate errors for given representations. Bronze: absolute error only; Silver: absolute and relative; Gold: error propagation in multi-step calculation.

Independent Practice (25 min): Research task: "Find a real-world example where floating-point error caused a significant problem." (e.g., Ariane 5, Patriot missile, Vancouver stock exchange). Students prepare 2-minute presentation.

Plenary (10 min): Mini-presentations (3 students). Class discussion: how could these errors have been prevented?

Resources

Normalisation practice worksheets

Error calculation problem sets

Case study articles: Ariane 5 explosion, Patriot missile failure

Python script demonstrating floating-point accumulation errors

Assessment Points

Formative: Exit ticket (Lesson 1) — normalisation

Formative: Error calculation worksheet (Lesson 2)

Homework: Normalise 6 numbers; calculate absolute and relative error for 4 representations; research one real-world floating-point error case

Bronze: Normalise simple positive numbers; calculate absolute error with guidance.

Silver: Normalise positive and negative numbers; calculate both absolute and relative error; explain underflow/overflow.

Gold: Handle complex normalisation; analyse error propagation; evaluate floating-point suitability for specific applications; present real-world case study.

Week 5: TCP/IP Protocol Stack

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Learning Objectives

Describe the four layers of the TCP/IP model: Application, Transport, Internet, Link

Explain the function of each layer and the protocols associated with it

Describe the encapsulation process as data moves down the stack

Compare TCP/IP with the OSI seven-layer model

Explain the role of ports in the transport layer

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): "How does an email get from your phone to a server in another country?" Brainstorm on whiteboard.

Direct Instruction (30 min): Present TCP/IP stack as a layered cake or postal system analogy. Each layer: Application (HTTP, FTP, SMTP, DNS), Transport (TCP, UDP), Internet (IP, ICMP), Link (Ethernet, Wi-Fi). Explain how each layer adds/removes headers.

Worked Example (15 min): Trace a web request through all four layers. Show headers added at each stage.

Guided Practice (25 min): Students complete a blank TCP/IP stack diagram, filling in layer names, protocols, and header information. Pairs check each other's work.

Plenary (10 min): Quick quiz: "Which layer handles IP addresses? Which handles reliability?"

Lesson 2 (2 hours):

Starter (10 min): Compare TCP/IP 4-layer model with OSI 7-layer model. Which layers correspond?

Direct Instruction (20 min): Deep dive into Transport layer: TCP vs UDP. TCP: connection-oriented, reliable, ordered, flow control, congestion control. UDP: connectionless, fast, no guarantees. Use cases for each.

Worked Example (15 min): Three-way handshake (SYN, SYN-ACK, ACK) diagrammed and explained. Sequence numbers and acknowledgements.

Guided Practice (25 min): Scenario cards: "Live video streaming" — TCP or UDP? Why? "Bank transaction" — TCP or UDP? Why? Students justify choices in pairs.

Independent Practice (30 min): Research and create a comparison table: TCP vs UDP across 8 criteria (connection, reliability, ordering, speed, overhead, use cases, header size, examples).

Plenary (10 min): Share comparison tables. Class vote on most surprising difference.

Resources

TCP/IP stack diagram posters for classroom walls

Wireshark packet capture demonstrations (sanitised)

OSI vs TCP/IP comparison handout

Video: "How the Internet Works in 5 Minutes" (Code.org)

Assessment Points

Formative: TCP/UDP scenario card justifications

Formative: Comparison table quality check

Homework: Draw complete encapsulation diagram for a DNS query; label all headers

Bronze: Name the four layers and one protocol per layer; identify TCP vs UDP for given scenarios.

Silver: Explain encapsulation process; compare TCP and UDP across multiple criteria; map TCP/IP to OSI.

Gold: Analyse packet capture data; explain three-way handshake in detail; evaluate when UDP might be preferable despite lack of reliability.

Week 6: Routing Algorithms

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Learning Objectives

Explain the purpose of routing in packet-switched networks

Describe distance-vector routing (RIP) and its limitations (count-to-infinity)

Describe link-state routing (OSPF) and its advantages

Trace Dijkstra's shortest-path algorithm on a network graph

Compare static and dynamic routing approaches

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): "What's the shortest route from school to the city centre?" Map discussion introduces graph concepts.

Direct Instruction (25 min): Routing as finding optimal paths in a graph. Introduce distance-vector: each router shares its entire routing table with neighbours. Bellman-Ford equation. Demonstrate with 4-router network.

Worked Example (20 min): Teacher traces routing table updates across 3 iterations. Show how count-to-infinity problem arises when a link fails.

Guided Practice (25 min): Students given a 5-router network. Calculate initial routing tables, then update after one link failure. Identify the count-to-infinity problem.

Plenary (10 min): Solutions reviewed. Split-ring horizon and poison reverse introduced as solutions.

Lesson 2 (2 hours):

Starter (10 min): Review distance-vector limitations. "Is there a better way?"

Direct Instruction (25 min): Link-state routing: each router knows entire network topology. Dijkstra's algorithm introduced step-by-step. Priority queue, tentative/permanent labels.

Worked Example (25 min): Teacher traces Dijkstra's algorithm on a 6-node network. Table format: node, shortest distance, previous node. Students follow on identical worksheet.

Guided Practice (25 min): Students apply Dijkstra's to a new 7-node network. Bronze: first 3 steps provided; Silver: first step provided; Gold: complete from scratch.

Independent Practice (20 min): Comparison essay plan: "Compare distance-vector and link-state routing for a university campus network." Bullet points only.

Plenary (15 min): Class discussion of comparison points. Teacher models structured comparison paragraph.

Resources

Network topology worksheets (printed A3)

Dijkstra's algorithm step-by-step template

Cisco Packet Tracer simulation files (if available)

Video: "Distance Vector vs Link State Routing" (NetworkChuck)

Assessment Points

Summative: Quiz 2 (Week 6) — Networks & Routing (20 marks)

Formative: Dijkstra's algorithm tracing worksheet

Homework: Complete Dijkstra's on a given network; write 3 comparison points between distance-vector and link-state

Bronze: Identify routing table entries; follow provided Dijkstra's steps; name one limitation of distance-vector.

Silver: Calculate routing tables for small networks; complete Dijkstra's algorithm with partial guidance; explain count-to-infinity.

Gold: Solve complex network routing problems; trace Dijkstra's from scratch; evaluate routing protocol choice for given scenarios; explain split-horizon and poison reverse.

Week 7: DNS Deep Dive

Week 7: Domain Name System (DNS)

Learning Objectives

Explain the hierarchical structure of DNS (root, TLD, authoritative, subdomain)

Describe the DNS resolution process: recursive and iterative queries

Identify common DNS record types: A, AAAA, CNAME, MX, NS, TXT, SOA

Explain the role of caching in DNS performance

Discuss DNS security concerns and DNSSEC

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): "What happens when you type `www.google.com`?" Collect initial ideas.

Direct Instruction (30 min): DNS as the "phonebook of the Internet." Hierarchical tree structure: root servers (.), TLD servers (.com, .org, .uk), authoritative servers (example.com). Recursive resolver role.

Worked Example (20 min): Teacher traces complete DNS resolution for `www.example.com`. Show recursive query from client to resolver, then iterative queries from resolver to root, TLD, and authoritative servers.

Guided Practice (25 min): Students trace resolution for `mail.vpa.edu.cn` on provided diagram. Label each query and response. Pairs compare traces.

Plenary (5 min): Exit ticket: "Why is DNS hierarchical rather than flat?"

Lesson 2 (2 hours):

Starter (10 min): Review DNS hierarchy. "What information does each server actually store?"

Direct Instruction (25 min): DNS record types: A (IPv4 address), AAAA (IPv6), CNAME (alias), MX (mail server), NS (name server), TXT (text), SOA (start of authority). Show real `dig/nslookup` output.

Worked Example (15 min): Interpret a zone file snippet. Identify record types and their purposes.

Guided Practice (25 min): Students given 5 scenarios ("A company wants email at `mail.example.com`" — which record?). Match scenarios to record types.

Independent Practice (25 min): Research task: "What is DNSSEC? How does it prevent DNS spoofing?" Students read provided article and summarise in 5 bullet points.

Plenary (10 min): Share DNSSEC summaries. Discuss: "Why isn't DNSSEC universally deployed?"

Resources

DNS hierarchy diagram (A1 poster)

Real `dig` command output samples

Article: "DNSSEC: What It Is and Why It Matters" (ICANN)

Online tool: <https://dns.google/> for live queries

Assessment Points

Formative: DNS trace diagram (Lesson 1)

Formative: Record type matching exercise

Homework: Use nslookup/dig to query 3 domains; record and interpret A, MX, and NS records

Bronze: Name the levels of DNS hierarchy; identify A and MX records.

Silver: Trace complete DNS resolution; identify all common record types; explain caching benefits.

Gold: Analyse real DNS data; explain DNSSEC operation; evaluate security trade-offs; discuss CDN implications for DNS.

Week 8: Email Protocols & Web Technologies

Week 8: Email Protocols, HTML & CSS Basics

Learning Objectives

Compare SMTP, POP3, and IMAP protocols for email transmission and retrieval

Explain the role of MIME in supporting non-ASCII email content

Describe HTTP request methods (GET, POST, PUT, DELETE) and response codes

Construct basic HTML documents using semantic tags

Apply CSS selectors and properties to style HTML elements

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): "How many emails did you send yesterday? What happened behind the scenes?"

Direct Instruction (25 min): Email architecture: MUA (Mail User Agent) → MSA → MTA → MDA → MUA. SMTP for sending (port 25/587). POP3 (port 110) vs IMAP (port 143) for retrieval. Compare: POP3 downloads and deletes; IMAP syncs across devices.

Worked Example (15 min): Trace an email from Alice@example.com to Bob@sample.org. Show SMTP handshake, envelope vs header, MIME for attachments.

Guided Practice (25 min): Comparison table: SMTP vs POP3 vs IMAP across 6 criteria (direction, port, persistence, sync, offline access, security). Pairs complete, then class verifies.

Plenary (5 min): Quick-fire: "Which protocol would you use on a phone? On a shared library computer?"

Lesson 2 (2 hours):

Starter (10 min): "What happens when you click a link?" Review HTTP from Week 5.

Direct Instruction (20 min): HTTP methods: GET (retrieve), POST (submit), PUT (update), DELETE (remove). Status codes: 1xx informational, 2xx success, 3xx redirect, 4xx client error, 5xx server error. Common codes: 200, 301, 404, 500.

Worked Example (10 min): Teacher live-codes a basic HTML page: , , , , semantic tags (, , ,).

Guided Practice (30 min): Students create a personal profile page in HTML. Bronze: basic structure with headings and paragraphs; Silver: add lists, links, and an image; Gold: add a table and form elements.

Independent Practice (30 min): Add CSS styling. Bronze: change colours and fonts; Silver: add class selectors, box model properties; Gold: responsive design with media queries, flexbox layout.

Plenary (10 min): Showcase 3 student pages. Peer feedback using rubric: structure, semantics, styling, creativity.

Resources

Email protocol comparison handout

HTTP status code reference card

CodePen.io accounts for HTML/CSS practice

MDN Web Docs: HTML and CSS references

Assessment Points

Formative: Protocol comparison table

Formative: HTML/CSS practical task marked against rubric

Homework: Create a 3-page website about a topic of choice; include navigation, styled with CSS

Bronze: Identify SMTP, POP3, IMAP purposes; create basic HTML page with headings and paragraphs.

Silver: Compare email protocols across criteria; create structured HTML with semantic tags and basic CSS styling.

Gold: Explain MIME and SMTP authentication; create responsive multi-page website with advanced CSS; explain HTTP caching and security headers.

Week 9: Consolidation & Formative Assessment

Week 9: Review, Practice & Formative Test

Learning Objectives

Consolidate understanding of Weeks 1–8 topics

Identify personal strengths and areas for improvement

Apply examination techniques to structured questions

Peer-teach challenging concepts to reinforce understanding

Lesson Activities

Lesson 1 (2 hours):

Starter (10 min): "One thing I understand well / One thing I need help with" — sticky note activity.

Direct Instruction (15 min): Teacher addresses top 3 areas of confusion from sticky notes (typically: K-map grouping, Dijkstra's algorithm, DNS resolution).

Guided Practice (45 min): Rotating revision stations: (1) Boolean algebra & K-maps, (2) Floating-point & normalisation, (3) TCP/IP & routing, (4) DNS & email. Students spend 10 minutes at each station with differentiated problem sets.

Peer Teaching (30 min): Students paired across ability levels. Each explains one concept to their partner. Teacher circulates with assessment checklist.

Plenary (10 min): "What will you revise tonight?" Personal revision plan written.

Lesson 2 (2 hours):

Formative Test (90 min): 40-mark paper covering Weeks 1–8. Mix of short answer, calculation, and extended response. Bilingual glossary provided.

Self-Assessment (15 min): Students mark their own work using provided mark scheme. Identify errors and misconceptions. Complete reflection sheet.

Target Setting (15 min): Students set 3 specific targets for improvement before mock exam. Teacher reviews and signs off.

Resources

Revision station problem sets (4 levels per station)

Formative test paper (40 marks, 90 minutes)

Mark scheme and reflection sheet

Personal revision plan template

Assessment Points

Formative: 40-mark test (Weeks 1-8) — diagnostic, not graded

Formative: Self-assessment and reflection sheet

Formative: Personal target-setting document

EAL Support — Week 9

Bilingual glossary for all technical terms used in test

Extra reading time (+10 minutes for 90-minute test)

Visual diagrams provided for all network questions

Sentence starters for extended-response questions

One-to-one review of targets with EAL coordinator

Cross-Curricular Links

Subject	Connection	Activity
Mathematics	Binary arithmetic, graph theory, error calculation	Joint lesson on Dijkstra's algorithm with Maths department
Physics	Electrical signals, logic gates, circuit design	Lab session building simple logic circuits
English	Technical writing, structured arguments	Essay practice: "Evaluate the security of DNS"
Business	Network infrastructure, e-commerce protocols	Case study: Amazon's global network architecture

Homework Schedule (Weeks 1 – 9)

Week	Homework Task	Estimated Time
1	Simplify 5 Boolean expressions; prove 2 using truth tables	45 min
2	Complete 5 K-map problems; verify one using Logisim	45 min

Week	Homework Task	Estimated Time
3	Convert 5 denary to IEEE 754; explain 0.1 + 0.2 problem	60 min
4	Normalise 6 numbers; calculate errors; research case study	60 min
5	Draw encapsulation diagram for DNS query; label headers	45 min
6	Complete Dijkstra's algorithm; write 3 comparison points	60 min
7	Use dig/nslookup on 3 domains; interpret A, MX, NS records	45 min
8	Create 3-page website with HTML/CSS	90 min
9	Personal revision: weakest topic from formative test	60 min

关键术语复习 / Key Term Revision:

Boolean algebra = 布尔代数 | Karnaugh map = 卡诺图 | Floating-point = 浮点数 | Normalisation = 规范化 | TCP/IP = 传输控制协议/网际协议 | Routing = 路由 | DNS = 域名系统 | SMTP = 简单邮件传输协议 | HTML = 超文本标记语言 | CSS = 层叠样式表

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Weeks 1-9 Scheme of Work v1.0 | Semester 1 2024-2025